

Algebra 1
Unit 11
Lesson 1 Homework

Name _____
Date _____
Period _____

1. In the equation $y = a(b)^x$, a represents the ☐ initial amount. ☐ growth factor. and b represents ☐ initial amount ☐ growth factor
2. When b is greater than 1, the equation models ☐ exponential growth. ☐ exponential decay.
3. When b is between 0 and 1, the equation models ☐ exponential growth. ☐ exponential decay.

For questions 4-8, identify if each representation of a function is an example of exponential growth or exponential decay. Then justify your selection.

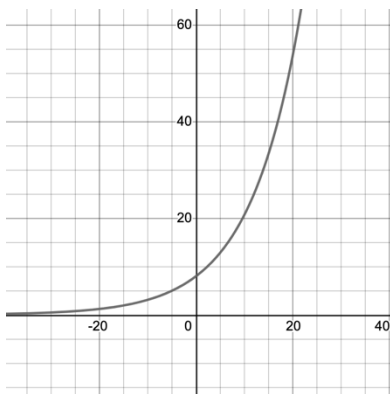
4.

x	y
-1	12
0	4
1	$\frac{4}{3}$
2	$\frac{4}{9}$
3	$\frac{4}{27}$

 represents _____ because...

5. $y = 25(0.79)^x$ represents _____ because...

6.



represents _____
because...

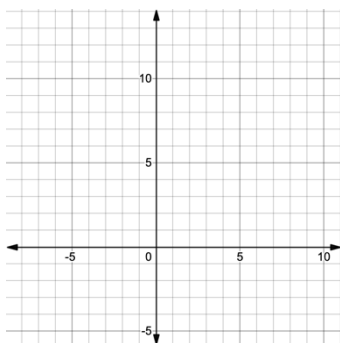
7. $y = 6(1 + 0.05)^x$ represents _____ because...

8.

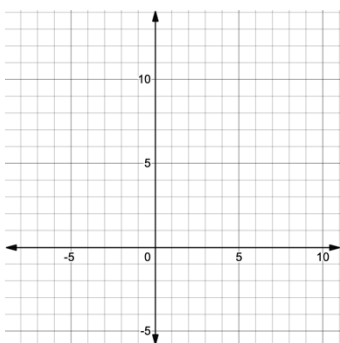
x	y
2	9
3	13.5
4	20.25
5	30.375

represents _____ because...

9. Sketch a graph that represents exponential growth.



10. Sketch a graph that represents exponential decay.



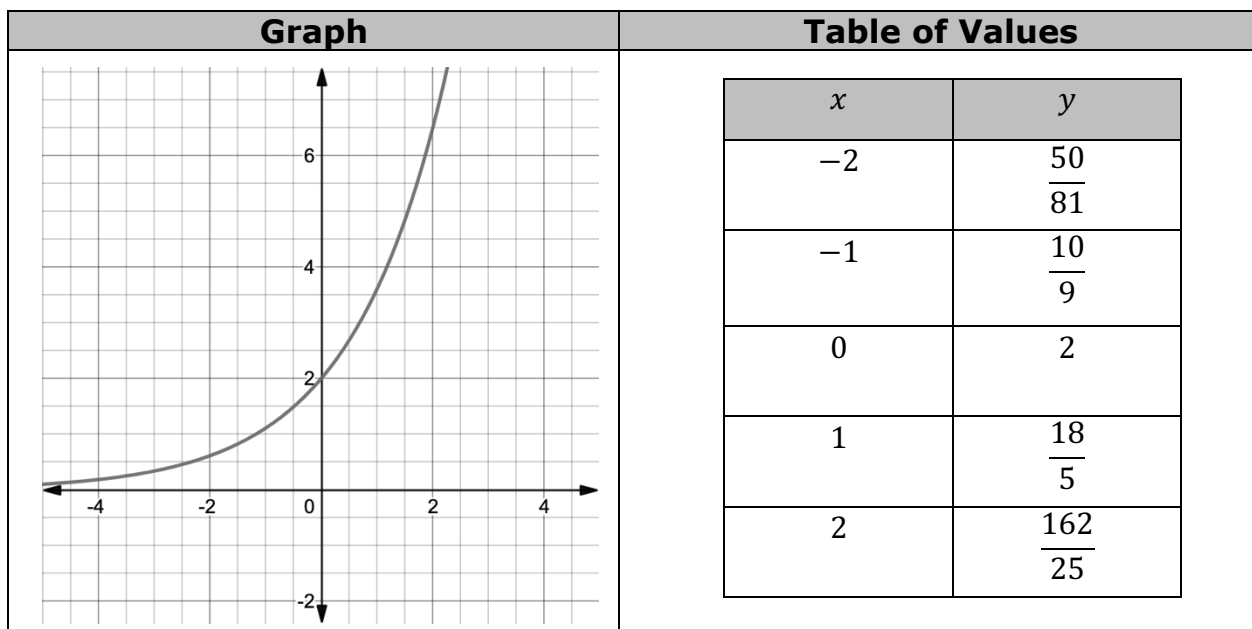
Algebra 1
Unit 11
Lesson 2 Homework

Name _____
Date _____
Period _____

1. The graph of $y = \left(\frac{5}{6}\right)^x$ is ☐ increasing
☐ decreasing and is ☐ positive.
☐ negative.

2. The graph of $y = 2(1 + 0.25)^x$ is ☐ increasing
☐ decreasing and is ☐ positive.
☐ negative.

Use the graph and table of $y = 2\left(\frac{9}{5}\right)^x$ to determine key features.



Key Features			
3. Domain		4. Range	
5. y - intercept		6. Constant Percent Rate of Change	
7. Increasing or Decreasing		8. Positive	
		Negative	
9. Left End Behavior		10. Right End Behavior	

Algebra 1

Unit 11

Lesson 3 Homework

Name _____

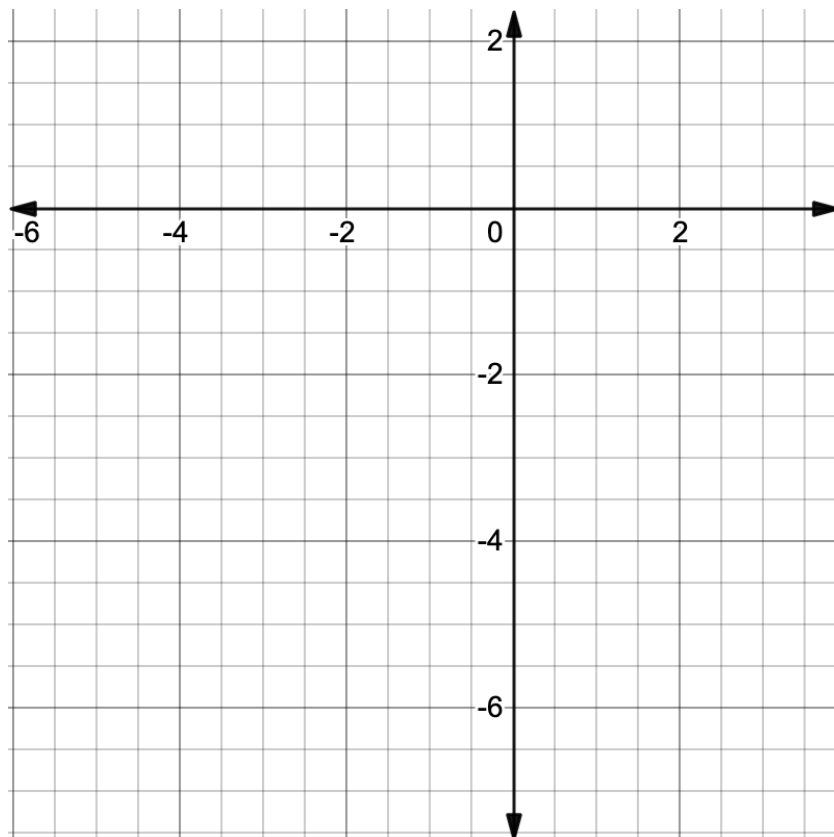
Date _____

Period _____

For questions 1 – 2, consider the function $f(x) = -4(1 + 0.65)^x$.

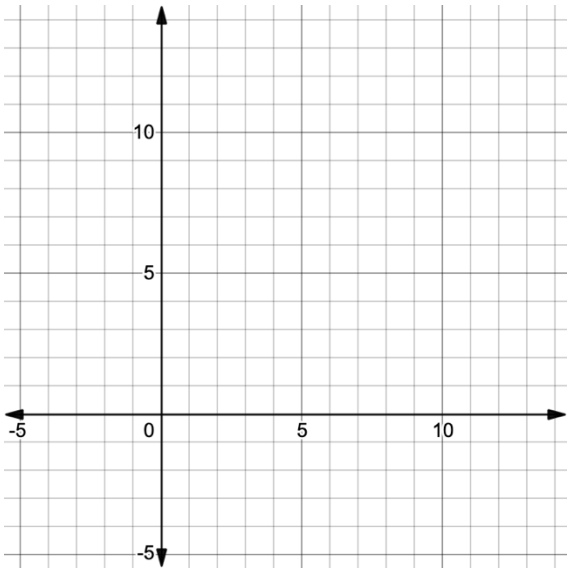
1. Since a is ☐ positive
☐ negative and b is ☐ between 0 and 1,
☐ greater than 1, the function
- is ☐ increasing.
☐ decreasing. The left end behavior of the graph is ☐ $y \rightarrow 0$
☐ $y \rightarrow -\infty$ and the
- right end behavior is ☐ $y \rightarrow 0$.
☐ $y \rightarrow -\infty$. The y -intercept of the function is ☐ -4 .
☐ 0.65 .
☐ 1.65 .

2. Sketch the graph of $f(x) = -4(1 + 0.65)^x$.



Graph the function of $y = 5\left(\frac{2}{3}\right)^x$, complete the table of values, and then determine all key features.

3. Graph



4. Table of Values

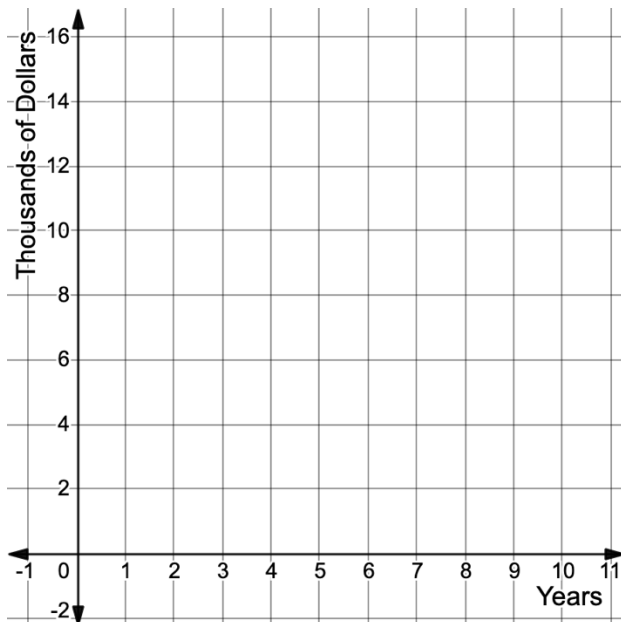
x	y
-2	
-1	
0	
1	
2	

Key Features	
5. Growth or Decay	
6. Domain and Range	
7. Increasing/Decreasing	
8. Constant Percent Rate of Change	
9. Left and Right End Behavior	
10. Positive or Negative	

Use the scenario below for questions 1 – 5.

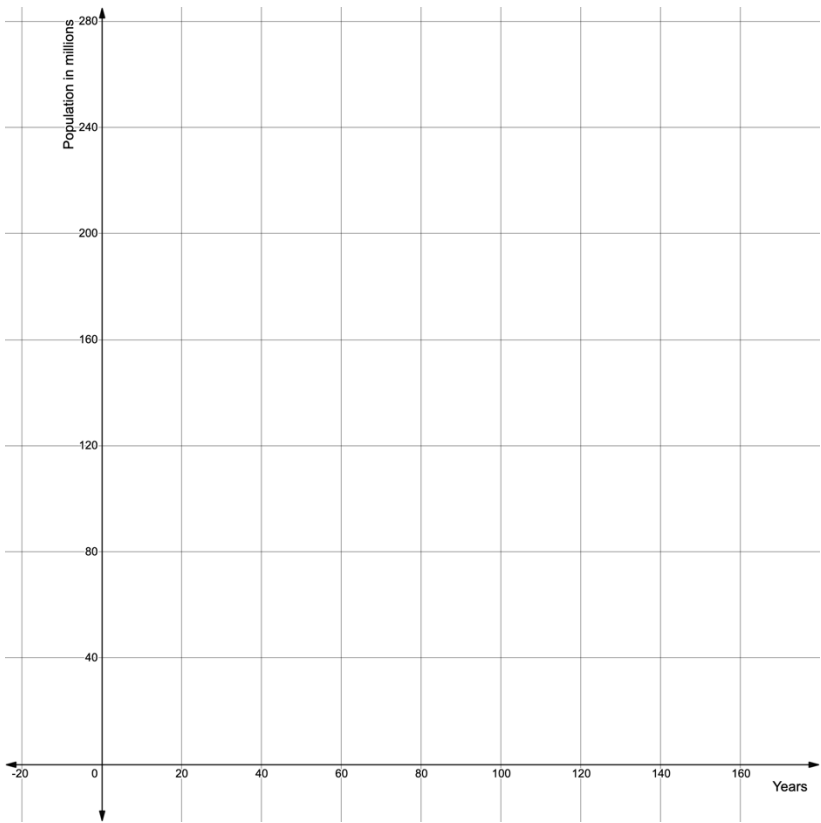
Tom purchased a used Honda CRV from a dealership for \$15,575. The car depreciates 15% per year. The situation can be modeled by the function $P(x) = \$15,575(0.85)^x$ where x represents the number of years since Tom purchased the car and P is the value of the car.

1. Does the equation $P(x) = \$15,575(0.85)^x$ represent exponential growth or decay?
2. What is the y -intercept and what does it represent in this situation?
3. Explain why the value of b is not 0.15, the rate of depreciation.
4. Graph the equation $P(x) = \$15,575(0.85)^x$.



The equation $f(t) = 76(1 + 0.013)^t$ represents the U.S. population and t represents the number of years after 1900. Graph the function below, then determine the key features and interpret how each relates to the context.

5.



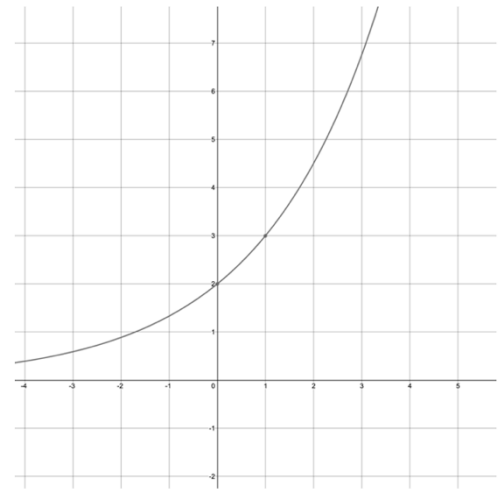
Key Features		Interpretations
6. Domain		
7. Range		
8. Y-intercept		
9. Constant Percent Rate of Change		
10. End Behavior		

The graph of an exponential function is shown.

1. Find the value of a by substituting the point $(0, 2)$ in for x and y in $y = a(b)^x$.

$$\boxed{} = a(b)^{\boxed{}}$$

2. Use the value of a found in question 1 and the point $(1, 3)$ to determine the value of b .



3. Write the function for the graph in both forms.

$f(x) = a(b)^x$	$f(x) = a(1 \pm r)^x$

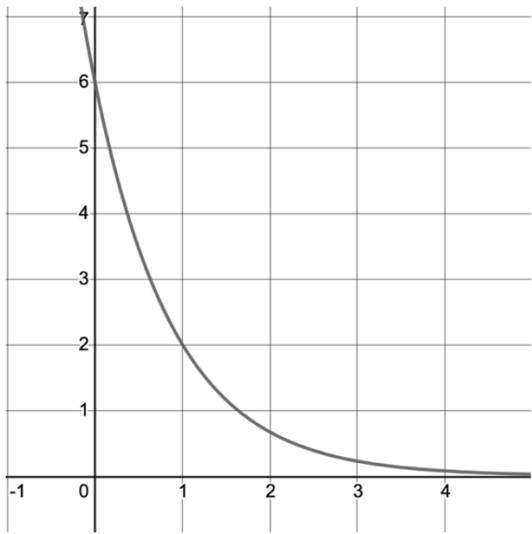
The table of an exponential function is shown.

x	0	1	2	3	4
$f(x)$	-7	-14	-28	-56	-112

4. What is the initial value, a ?
5. How does the value of $f(x)$ change as the x -value increases by 1?
6. Write the function in the form $f(x) = a(b)^x$

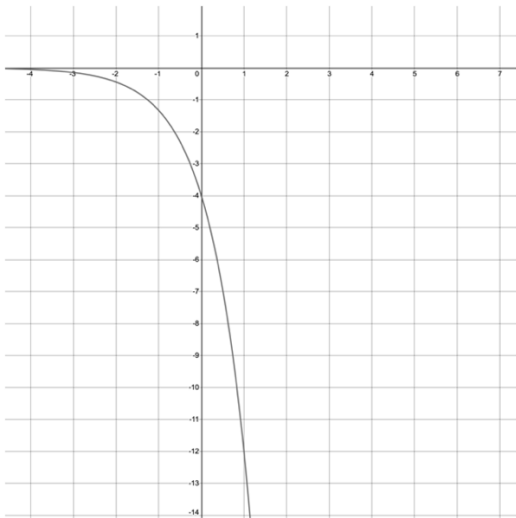
For questions 7 – 10, write an equation from the table or graph by finding the initial value, a , and common ratio, b .

7.



Function

8.



Function

9.

x	y
0	-5
1	$-\frac{25}{4}$
2	$-\frac{125}{16}$
3	$-\frac{625}{64}$
4	$-\frac{3125}{256}$

Function

10.

x	y
0	3
1	2.1
2	1.47
3	1.029
4	0.7203

Function

Use the situation below to answer questions 1 – 3.

A new Microsoft Surface computer costs \$1275, but depreciates 16.5% each year as new technology comes out.

1. Is this situation exponential growth or decay?
2. Two students, Josh and Eve were debating how to represent a function to model the value, V , after x years. Both of their functions are below. Explain who you agree with and why.

Josh	Eve
$V(x) = 1275(0.835)^x$	$V(x) = 1275(1.165)^x$

3. Write the correct function that models the context using both forms.

$V(x) = a(b)^x$	$V(x) = a(1 \pm r)^x$

Use the situation below to answer questions 4 – 6.

A new school opened with 225 students in 2021 and plans to increase by 13.3% per year until they reach full capacity.

4. Is this situation exponential growth or decay?
5. Write an equation that models the population of the school, P , after x years since the opening of the school in 2021.

6. Write the correct function that models the context using both forms.

$P(x) = a(b)^x$	$P(x) = a(1 \pm r)^x$

7. A three-bedroom house in the suburbs of Jacksonville sold for \$320,628. If the housing prices are expected to increase 2.8% annually, write a function that models the cost of a house, C , x years after it was sold.
8. A small local company reported a profit of \$175,000 last year, but is struggling due to the pandemic. They are reporting a decrease of 13.6% each year. Write a function that represents the profit, P , after t years.
9. As of 2019, the world population is 8.215 billion people and is expected to grow at a rate of 1.2% per year. Write an equation to model the population growth, where $P(y)$ is the population in billions of people and y is the time in years.
10. A culture of 22 bacteria grows at a rate of 150% each hour. Tyler wrote the equation $B(t) = 22(1.5)^t$ to represent the population of the bacteria after t hours. Describe and correct the error.